

Week 7 — While Loops & The Guessing Game

Name: ____ Date: ____

A *while* loop repeats **until its condition fails**. No update line → it never stops!

Today's words

Word	What it means
while loop	repeats as long as its condition holds
condition	the yes/no check (1 or 0) before every lap
update line	the line inside that changes the variable — the escape route
infinite loop	a loop that never stops (escape hatch: Ctrl+C)
binary search	always guess the middle — half the suspects vanish each time

1 • Be the computer: trace the loop 🤖

```
n = 6;
while n > 0
    disp(n)
    n = n - 2;
end
disp('done')
```

lap	what prints	n after the update
1		
2		
3		

Last line on the screen: ____

Danger question: if the line `n = n - 2;` were deleted, what would happen?

2 • The halving chain 🧵

Each higher/lower answer cuts the suspects in half (round down). Fill in the chain:

100 → 50 → 25 → _ → _ → ____ → 1

Count the numbers in the chain: _ — **that's why** _ guesses always win. Check: $2^7 = \underline{\hspace{1cm}}$, which is bigger than 100. ✓

3 • Binary search on paper 🔍

A friend picks **11**, from 1 to 16. You always guess the **middle** of what's left (round down). Finish the hunt — row 1 is done for you:

numbers still possible	your guess	friend says
1-16	8	higher!
9-16		
		correct! 🎉

🧠 Brain teaser (optional — take it home)

Seven guesses crack 1-100 because $2^7 = 128 > 100$. **How many guesses would you need for 1 to 1,000? For 1 to 1,000,000?** (Hint: keep doubling — 2, 4, 8, 16... — and count how many doublings pass each target.) Then the mind-bender: going from 1,000 to 1,000,000 multiplies the suspects by a THOUSAND — how many *extra* guesses did it cost? Bring your answer next week for a shout-out.